

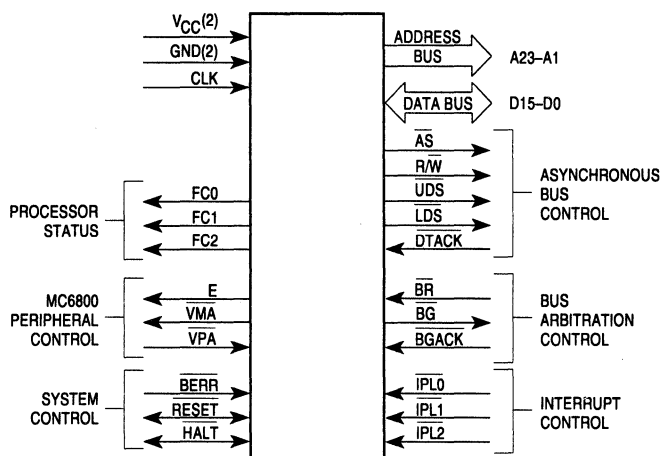
SECTION 3

SIGNAL DESCRIPTION

This section contains descriptions of the input and output signals. The input and output signals can be functionally organized into the groups shown in Figure 3-1 (for the MC68000, the MC68HC000 and the MC68010), Figure 3-2 (for the MC68HC001), Figure 3-3 (for the MC68EC000), Figure 3-4 (for the MC68008, 48-pin version), and Figure 3-5 (for the MC68008, 52-pin version). The following paragraphs provide brief descriptions of the signals and references (where applicable) to other paragraphs that contain more information about the signals.

NOTE

The terms **assertion** and **negation** are used extensively in this manual to avoid confusion when describing a mixture of "active-low" and "active-high" signals. The term assert or assertion is used to indicate that a signal is active or true, independently of whether that level is represented by a high or low voltage. The term negate or negation is used to indicate that a signal is inactive or false.



**Figure 3-1. Input and Output Signals
(MC68000, MC68HC000 and MC68010)**

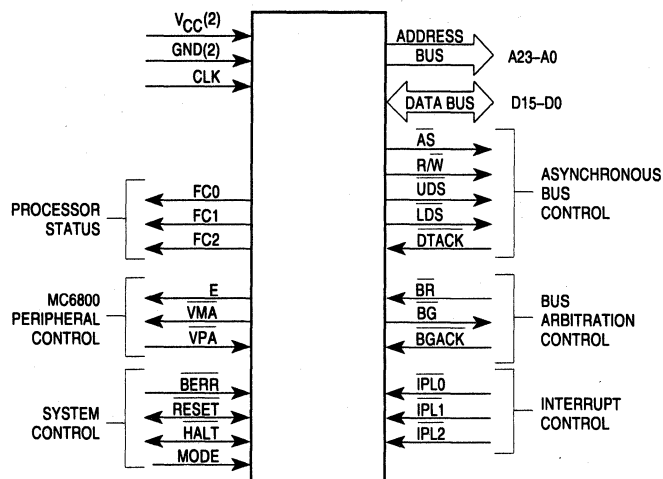


Figure 3-2. Input and Output Signals (MC68HC001)

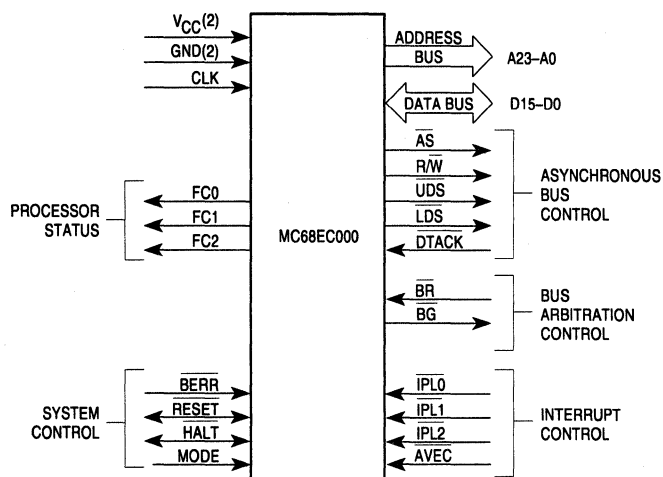


Figure 3-3. Input and Output Signals (MC68EC000)

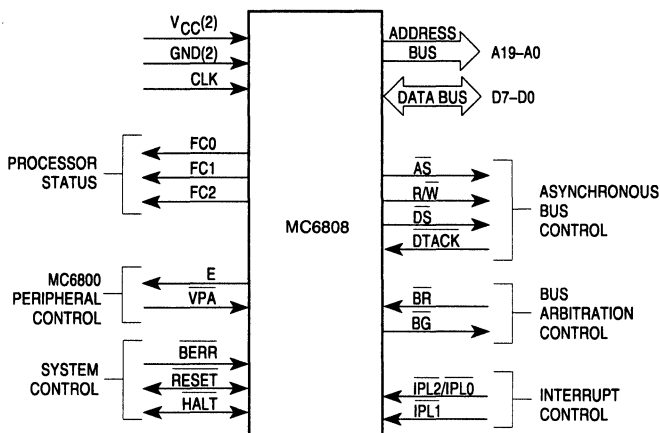


Figure 3-4. Input and Output Signals (MC68008, 48-Pin Version)

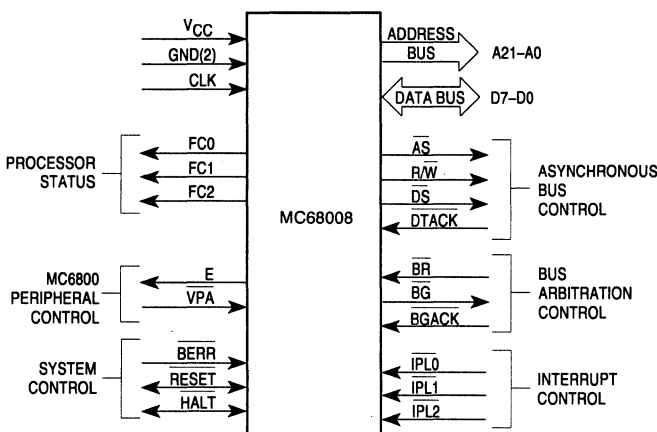


Figure 3-5. Input and Output Signals (MC68008, 52-Pin Version)

3.1 ADDRESS BUS (A23-A1)

This 23-bit, unidirectional, three-state bus is capable of addressing 16 Mbytes of data. This bus provides the address for bus operation during all cycles except interrupt acknowledge cycles and breakpoint cycles. During interrupt acknowledge cycles, address lines A1, A2, and A3 provide the level number of the interrupt being acknowledged, and address lines A23-A4 are driven to logic high.

Address Bus (A23–A0)

This 24-bit, unidirectional, three-state bus is capable of addressing 16 Mbytes of data. This bus provides the address for bus operation during all cycles except interrupt acknowledge cycles and breakpoint cycles. During interrupt acknowledge cycles, address lines A1, A2, and A3 provide the level number of the interrupt being acknowledged, and address lines A23–A4 and A0 are driven to logic high. In 16-Bit mode, A0 is always driven high.

MC68008 Address Bus

The unidirectional, three-state buses in the two versions of the **MC68008** differ from each other and from the other processor bus only in the number of address lines and the addressing range. The 20-bit address (A19–A0) of the 48-pin version provides a 1-Mbyte address space; the 52-pin version supports a 22-bit address (A21–A0), extending the address space to 4 Mbytes. During an interrupt acknowledge cycle, the interrupt level number is placed on lines A1, A2, and A3. Lines A0 and A4 through the most significant address line are driven to logic high.

3.2 DATA BUS (D15–D0; MC68008: D7–D0)

This bidirectional, three-state bus is the general-purpose data path. It is 16 bits wide in all the processors except the **MC68008** which is 8 bits wide. The bus can transfer and accept data of either word or byte length. During an interrupt acknowledge cycle, the external device supplies the vector number on data lines D7–D0. The MC68EC000 and MC68HC001 use D7–D0 in 8-bit mode, and D15–D8 are undefined.

3.3 ASYNCHRONOUS BUS CONTROL

Asynchronous data transfers are controlled by the following signals: address strobe, read/write, upper and lower data strobes, and data transfer acknowledge. These signals are described in the following paragraphs.

Address Strobe ($\overline{AS}$).

This three-state signal indicates that the information on the address bus is a valid address.

Read/Write ($R/\overline{W}$).

This three-state signal defines the data bus transfer as a read or write cycle. The $R/\overline{W}$ signal relates to the data strobe signals described in the following paragraphs.

Upper And Lower Data Strobes ($\overline{UDS}$, $\overline{LDS}$).

These three-state signals and $R/\overline{W}$ control the flow of data on the data bus. Table 3-1 lists the combinations of these signals and the corresponding data on the bus. When the $R/\overline{W}$ line is high, the processor reads from the data bus. When the $R/\overline{W}$ line is low, the processor drives the data bus. In 8-bit mode, $\overline{UDS}$ is always forced high and the $\overline{LDS}$ signal is used.

Table 3-1. Data Strobe Control of Data Bus

UDS	LDS	R/W	D8–D15	D0–D7
High	High	—	No Valid Data	No Valid Data
Low	Low	High	Valid Data Bits 15–8	Valid Data Bits 7–0
High	Low	High	No Valid Data	Valid Data Bits 7–0
Low	High	High	Valid Data Bus 15–8	No Valid Data
Low	Low	Low	Valid Data Bits 15–8	Valid Data Bits 7–0
High	Low	Low	Valid Data Bits 7–0*	Valid Data Bits 7–0
Low	High	Low	Valid Data Bits 15–8	Valid Data Bits 15–8*

*These conditions are a result of current implementation and may not appear on future devices.

Data Strobe ($\overline{DS}$) (MC68008)

This three-state signal and $R/\overline{W}$ control the flow of data on the data bus of the **MC68008**. Table 3-2 lists the combinations of these signals and the corresponding data on the bus. When the $R/\overline{W}$ line is high, the processor reads from the data bus. When the $R/\overline{W}$ line is low, the processor drives the data bus.

**Table 3-2. Data Strobe Control
of Data Bus (MC68008)**

DS	R/W	D0–D7
1	—	No Valid Data
0	1	Valid Data Bits 7–0 (Read Cycle)
0	0	Valid Data Bits 7–0 (Write Cycle)

Data Transfer Acknowledge ($\overline{DTACK}$).

This input signal indicates the completion of the data transfer. When the processor recognizes $\overline{DTACK}$ during a read cycle, data is latched, and the bus cycle is terminated. When $\overline{DTACK}$ is recognized during a write cycle, the bus cycle is terminated.

3.4 BUS ARBITRATION CONTROL

The bus request, bus grant, and bus grant acknowledge signals form a bus arbitration circuit to determine which device becomes the bus master device. In the 48-pin version of the MC68008 and MC68EC000, no pin is available for the bus grant acknowledge signal; this microprocessor uses a two-wire bus arbitration scheme. All M68000 processors can use two-wire bus arbitration.

Bus Request ($\overline{BR}$).

This input can be wire-ORed with bus request signals from all other devices that could be bus masters. This signal indicates to the processor that some other device needs to become the bus master. Bus requests can be issued at any time during a cycle or between cycles.

Bus Grant ($\overline{BG}$).

This output signal indicates to all other potential bus master devices that the processor will relinquish bus control at the end of the current bus cycle.

Bus Grant Acknowledge ($\overline{BGACK}$).

This input indicates that some other device has become the bus master. This signal should not be asserted until the following conditions are met:

1. A bus grant has been received.
2. Address strobe is inactive, which indicates that the microprocessor is not using the bus.
3. Data transfer acknowledge is inactive, which indicates that neither memory nor peripherals are using the bus.
4. Bus grant acknowledge is inactive, which indicates that no other device is still claiming bus mastership.

The 48-pin version of the **MC68008** has no pin available for the bus grant acknowledge signal and uses a two-wire bus arbitration scheme instead. If another device in a system supplies a bus grant acknowledge signal, the bus request input signal to the processor should be asserted when either the bus request or the bus grant acknowledge from that device is asserted.

3.5 INTERRUPT CONTROL ($\overline{IPL0}$, $\overline{IPL1}$, $\overline{IPL2}$)

These input signals indicate the encoded priority level of the device requesting an interrupt. Level seven, which cannot be masked, has the highest priority; level zero indicates that no interrupts are requested. $\overline{IPL0}$ is the least significant bit of the encoded level, and $\overline{IPL2}$ is the most significant bit. For each interrupt request, these signals must remain asserted until the processor signals interrupt acknowledge (FC2–FC0 and A19–A16 high) for that request to ensure that the interrupt is recognized.

NOTE

The 48-pin version of the **MC68008** has only two interrupt control signals: $\overline{IPL0}/\overline{IPL2}$ and $\overline{IPL1}$. $\overline{IPL0}/\overline{IPL2}$ is internally connected to both $\overline{IPL0}$ and $\overline{IPL2}$, which provides four interrupt priority levels: levels 0, 2, 5, and 7. In all other respects, the interrupt priority levels in this version of the **MC68008** are identical to those levels in the other microprocessors described in this manual.

3.6 SYSTEM CONTROL

The system control inputs are used to reset the processor, to halt the processor, and to signal a bus error to the processor. The outputs reset the external devices in the system and signal a processor error halt to those devices. The three system control signals are described in the following paragraphs.

Bus Error ($\overline{\text{BERR}}$)

This input signal indicates a problem in the current bus cycle. The problem may be the following:

1. No response from a device.
2. No interrupt vector number returned.
3. An illegal access request rejected by a memory management unit.
4. Some other application-dependent error.

Either the processor retries the bus cycle or performs exception processing, as determined by interaction between the bus error signal and the halt signal.

Reset ($\overline{\text{RESET}}$)

The external assertion of this bidirectional signal along with the assertion of $\overline{\text{HALT}}$ starts a system initialization sequence by resetting the processor. The processor assertion of $\overline{\text{RESET}}$ (from executing a RESET instruction) resets all external devices of a system without affecting the internal state of the processor. To reset both the processor and the external devices, the $\overline{\text{RESET}}$ and $\overline{\text{HALT}}$ input signals must be asserted at the same time.

Halt ($\overline{\text{HALT}}$)

An input to this bidirectional signal causes the processor to stop bus activity at the completion of the current bus cycle. This operation places all control signals in the inactive state and places all three-state lines in the high-impedance state (refer to Table 3-4).

When the processor has stopped executing instructions (in the case of a double bus fault condition, for example), the $\overline{\text{HALT}}$ line is driven by the processor to indicate the condition to external devices.

Mode (MODE) (MC68HC001/68EC000)

The MODE input selects between the 8-bit and 16-bit operating modes. If this input is grounded at reset, the processor will come out of reset in the 8-bit mode. If this input is tied high or floating at reset, the processor will come out of reset in the 16-bit mode. This input should be changed only at reset and must be stable two clocks after RESET is negated. Changing this input during normal operation may produce unpredictable results.

3.7 M6800 PERIPHERAL CONTROL

These control signals are used to interface the asynchronous M68000 processors with the synchronous M6800 peripheral devices. These signals are described in the following paragraphs.

Enable (E)

This signal is the standard enable signal common to all M6800 Family peripheral devices. A single period of clock E consists of 10 MC68000 clock periods (six clocks low, four clocks high). This signal is generated by an internal ring counter that may come up in any state. (At power-on, it is impossible to guarantee phase relationship of E to CLK.) The E signal is a free-running clock that runs regardless of the state of the MPU bus.

Valid Peripheral Address ($\overline{\text{VPA}}$)

This input signal indicates that the device or memory area addressed is an M6800 Family device or a memory area assigned to M6800 Family devices and that data transfer should be synchronized with the E signal. This input also indicates that the processor should use automatic vectoring for an interrupt. Refer to **Appendix B M6800 Peripheral Interface**.

Valid Memory Address ($\overline{\text{VMA}}$)

This output signal indicates to M6800 peripheral devices that the address on the address bus is valid and that the processor is synchronized to the E signal. This signal only responds to a $\overline{\text{VPA}}$ input that identifies an M6800 Family device.

The **MC68008** does not supply a $\overline{\text{VMA}}$ signal. This signal can be produced by a transistor-to-transistor logic (TTL) circuit; an example is described in **Appendix B M6800 Peripheral Interface**.

3.8 PROCESSOR FUNCTION CODES (FC0, FC1, FC2)

These function code outputs indicate the mode (user or supervisor) and the address space type currently being accessed, as shown in Table 3-3. The function code outputs are valid whenever $\overline{\text{AS}}$ is active.

Table 3-3. Function Code Outputs

Function Code Output			Address Space Type
FC2	FC1	FC0	
Low	Low	Low	(Undefined, Reserved)
Low	Low	High	User Data
Low	High	Low	User Program
Low	High	High	(Undefined, Reserved)
High	Low	Low	(Undefined, Reserved)
High	Low	High	Supervisor Data
High	High	Low	Supervisor Program
High	High	High	CPU Space

3.9 CLOCK (CLK)

The clock input is a TTL-compatible signal that is internally buffered for development of the internal clocks needed by the processor. This clock signal is a constant frequency square wave that requires no stretching or shaping. The clock input should not be gated off at any time, and the clock signal must conform to minimum and maximum pulse-width times listed in **Section 10 Electrical Characteristics**.

3.10 POWER SUPPLY (V_{CC} and GND)

Power is supplied to the processor using these connections. The positive output of the power supply is connected to the V_{CC} pins and ground is connected to the GND pins.

3.11 SIGNAL SUMMARY

Table 3-4 summarizes the signals discussed in the preceding paragraphs.

Table 3-4. Signal Summary

Signal Name	Mnemonic	Input/Output	Active State	HI-Z	
				On HALT	On Bus Relinquish
Address Bus	A0-A23	Output	High	Yes	Yes
Data Bus	D0-D15	Input/Output	High	Yes	Yes
Address Strobe	$\overline{AS}$	Output	Low	No	Yes
Read/Write	R/W	Output	Read-High Write-Low	No	Yes
Data Strobe	$\overline{DS}$	Output	Low	No	Yes
Upper and Lower Data Strobes	$\overline{UDS}$, $\overline{LDS}$	Output	Low	No	Yes
Data Transfer Acknowledge	$\overline{DTACK}$	Input	Low	No	No
Bus Request	$\overline{BR}$	Input	Low	No	No
Bus Grant	$\overline{BG}$	Output	Low	No	No
Bus Grant Acknowledge	$\overline{BGACK}$	Input	Low	No	No
Interrupt Priority Level	IPL0, IPL1, IPL2	Input	Low	No	No
Bus Error	$\overline{BERR}$	Input	Low	No	No
Mode	MODE	Input	High	—	—
Reset	$\overline{RESET}$	Input/Output	Low	No*	No*
Halt	$\overline{HALT}$	Input/Output	Low	No*	No*
Enable	E	Output	High	No	No
Valid Memory Address	VMA	Output	Low	No	Yes
Valid Peripheral Address	VPA	Input	Low	No	No
Function Code Output	FC0, FC1, FC2	Output	High	No	Yes
Clock	CLK	Input	High	No	No
Power Input	V _{CC}	Input	—	—	—
Ground	GND	Input	—	—	—

*Open drain.